

CS 6375

ASSIGNMENT 1

Names of students in your group:

Shisir Humagain

Number of free late days used: __0__

Note: You are allowed a **total** of 4 free late days for the **entire semester**. You can use at most 2 for each assignment. After that, there will be a penalty of 10% for each late day.

Please list clearly all the sources/references that you have used in this assignment.

- Google Gemini for Gradient Descent code

1. Introduction

The objective of this assignment was to implement linear regression using gradient descent and compare it with Scikit-Learn's SGDRegressor implementation. The Auto MPG dataset from the UCI Machine Learning Repository was selected because it is a regression dataset with both numerical and categorical features.

2. Dataset Description

Dataset: Auto MPG

Target Variable:

- MPG (Miles Per Gallon)

Number of Samples:

- 392 records after preprocessing

Features:

- Cylinders
 - Displacement
 - Horsepower
 - Weight
 - Acceleration
 - Model Year
 - Origin
-

3. Data Preprocessing

The following preprocessing steps were performed:

1. Removed duplicate records.
 2. Removed missing values.
 3. Converted categorical variables into numerical values.
 4. Standardized all feature variables using StandardScaler.
 5. Split data into 80% training and 20% testing sets.
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4. Part 1: Custom Gradient Descent Implementation

Algorithm

Linear Regression was implemented from scratch using vectorized gradient descent.

Weight Update Equation:

$$W = W - \alpha \nabla J(W)$$

Cost Function:

$$J(W) = (1/2m) \sum (h(x) - y)^2$$

Hyperparameter Tuning

The following combinations were tested:

Learning Rates:

- 0.001
- 0.005
- 0.01
- 0.05

Iterations:

- 1000
- 3000
- 5000

Results for all trials were stored in:

gradient_descent_tuning_log.csv

Best Parameters

Learning Rate: 0.05

Iterations: 10000

Best Training MSE: 10.979255842373474

Test Results

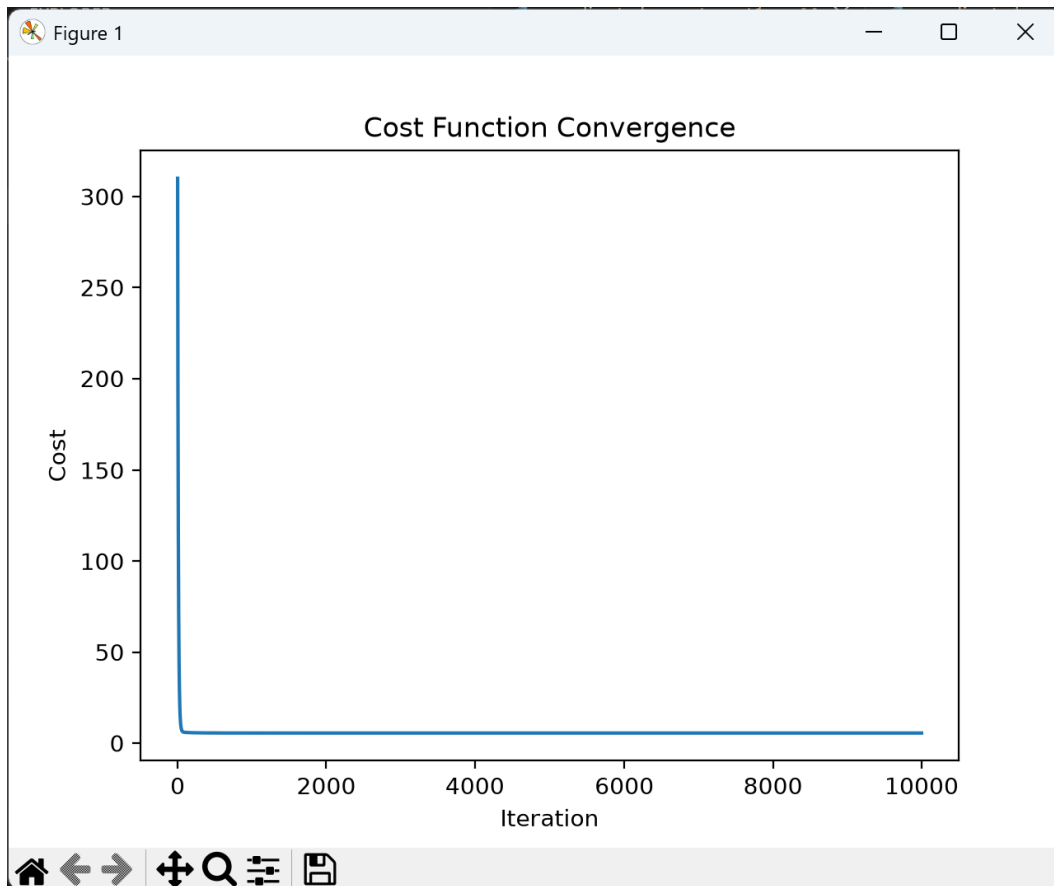
12	0.050	1000	10.989486
13	0.050	3000	10.979267
14	0.050	5000	10.979256
15	0.050	10000	10.979256
16	0.100	1000	10.979572

17	0.100	3000	10.979256
18	0.100	5000	10.979256
19	0.100	10000	10.979256

Test Results

MSE: 10.710864449504816

RMSE: 3.2727457049860775



5. Part 2: SGDRegressor Implementation

Model

SGDRegressor from Scikit-Learn was used to perform linear regression using stochastic gradient descent.

Hyperparameter Tuning

Parameters Tested:

Learning Rate:

- 0.0001
- 0.001
- 0.01
- 0.05

Iterations:

- 1000
- 3000
- 5000

Alpha:

- 0.0001
- 0.001

Results were saved in:

sgd_regressor_tuning_log.csv

Best Model Results

MSE: 9.7522

RMSE: 3.1228

MAE: 2.2231

R^2 Score: 0.8089

Explained Variance: 0.8090

Coefficients

Feature 0: 1.2919

Feature 1: -0.2579

Feature 2: -0.7034

Feature 3: -4.8318

Feature 4: -0.0910

Feature 5: 2.6249

Feature 6: 1.0747

Intercept: 23.1497

Interpretation

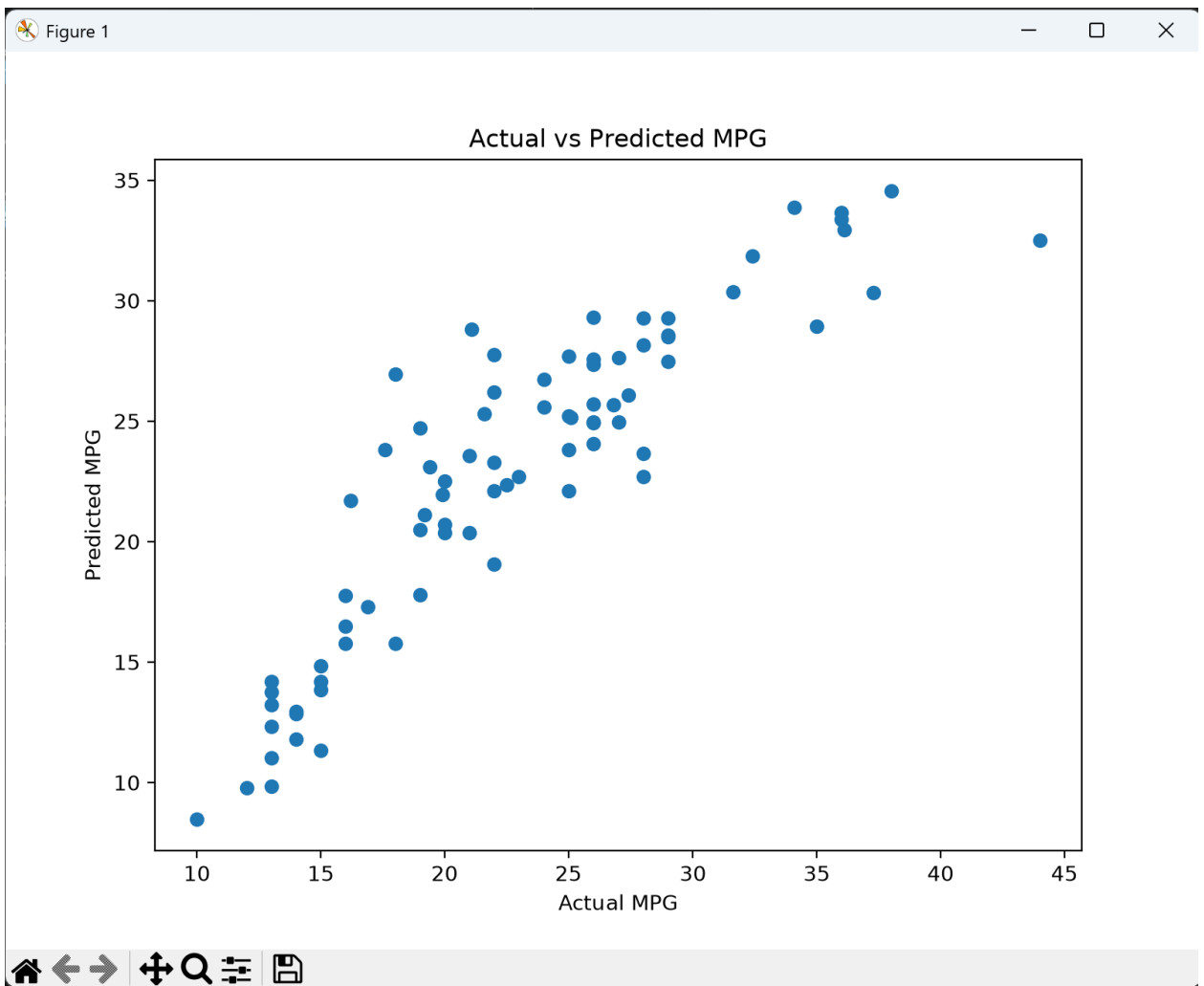
The model achieved an R^2 score of 0.8089, indicating that approximately 80.89% of the variance in MPG is explained by the model. Weight was identified as the most influential feature with a strong negative coefficient. Model year showed a positive impact on fuel efficiency, indicating newer vehicles tend to achieve higher MPG values.

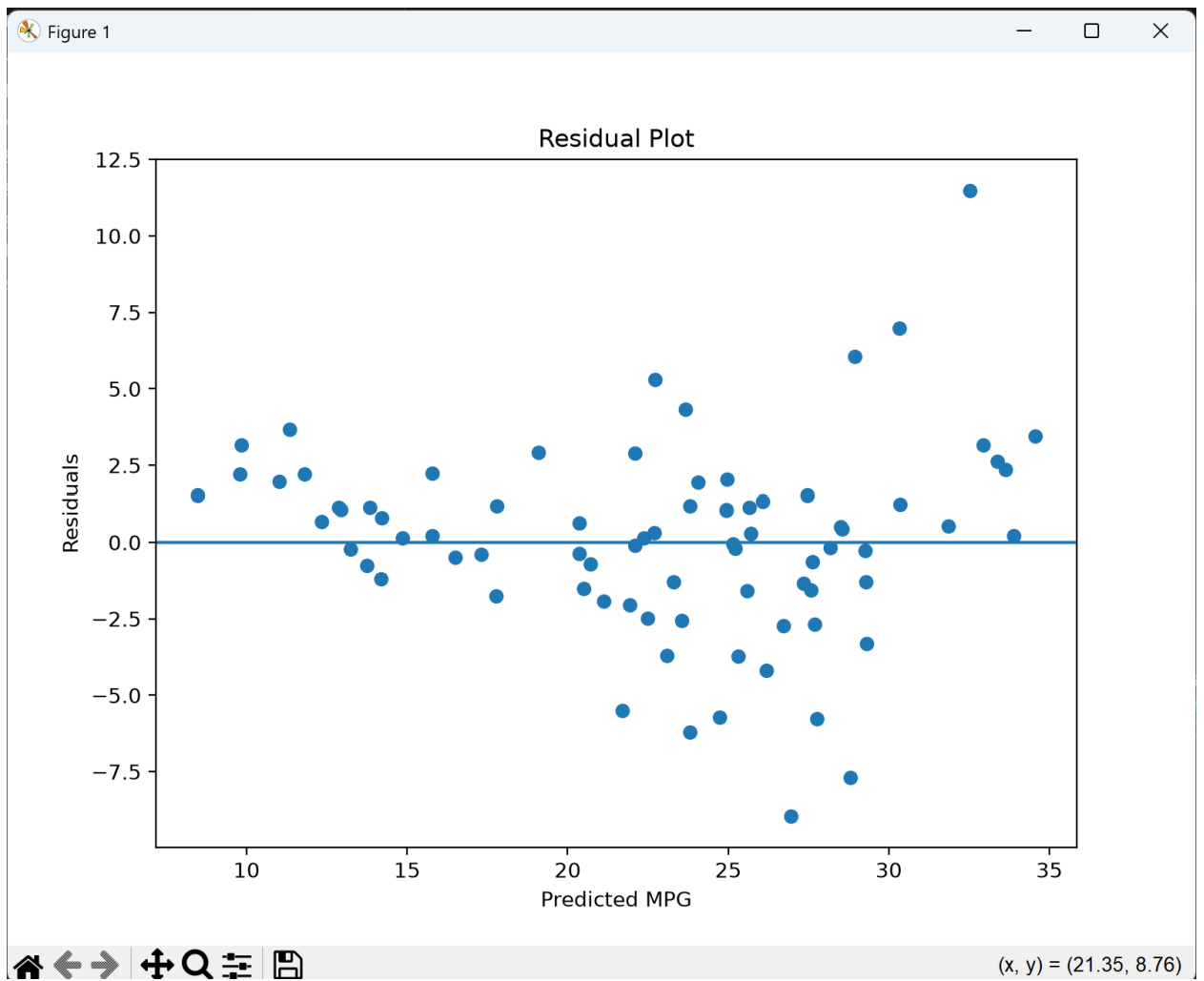
6. Evaluation Metrics

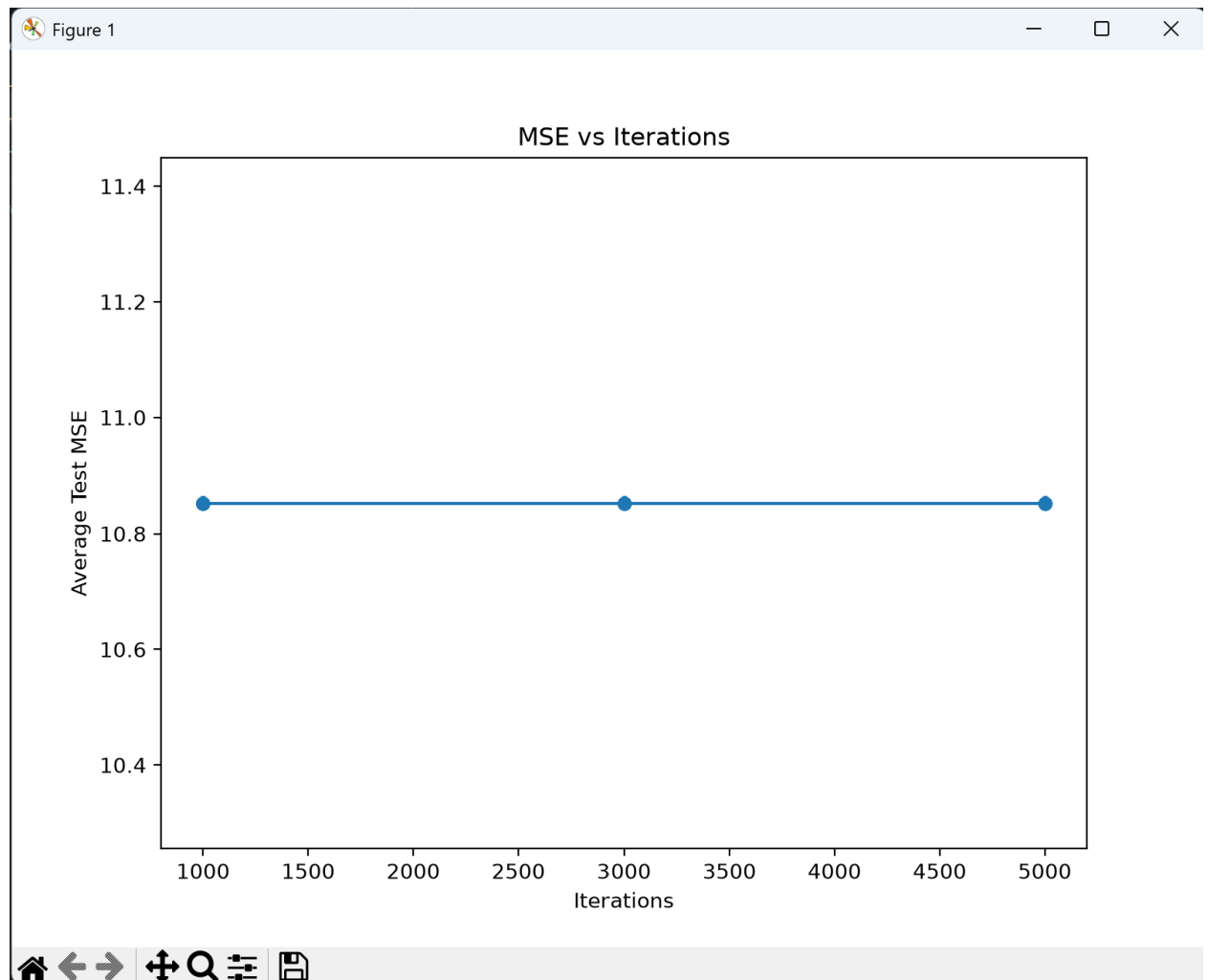
The following metrics were evaluated:

- Mean Squared Error (MSE)
 - Root Mean Squared Error (RMSE)
 - Mean Absolute Error (MAE)
 - R^2 Score
 - Explained Variance
 - Feature Coefficients
-

6. Plots







8. Conclusion

Both approaches successfully modeled the Auto MPG dataset. The custom implementation demonstrated the mechanics of gradient descent, while SGDRegressor provided a robust and optimized implementation. The best SGDRegressor model achieved an R^2 score of 0.8089, indicating strong predictive performance on the test dataset.